

Query - Can you show me **where to be careful and not make incisions?**

Response -

Model: Segment-Video

Prompt: Segment the internal carotid arteries and optic nerves.

Query - Identify if I'm prepared to transition **to tumor excision** with the **pituitary rongeur?**

Response -

Step 1:

Model: Trak-Instrument

Prompt: Track the location of bipolar forceps in the pituitary rongeur.

Step 2:

Model: Surgical-VQA

Prompt: Confirm readiness for the tumor excision and advise on final preperation

Query - I need **MRI-guided** overlaying of **the sella and right ICA** onto the surgical field

Response -

Step 1:

Model: Segment-MRI

Prompt: Segment sella and right internal carotid artery from MRI.

Step 2:

Model: Segment-Video

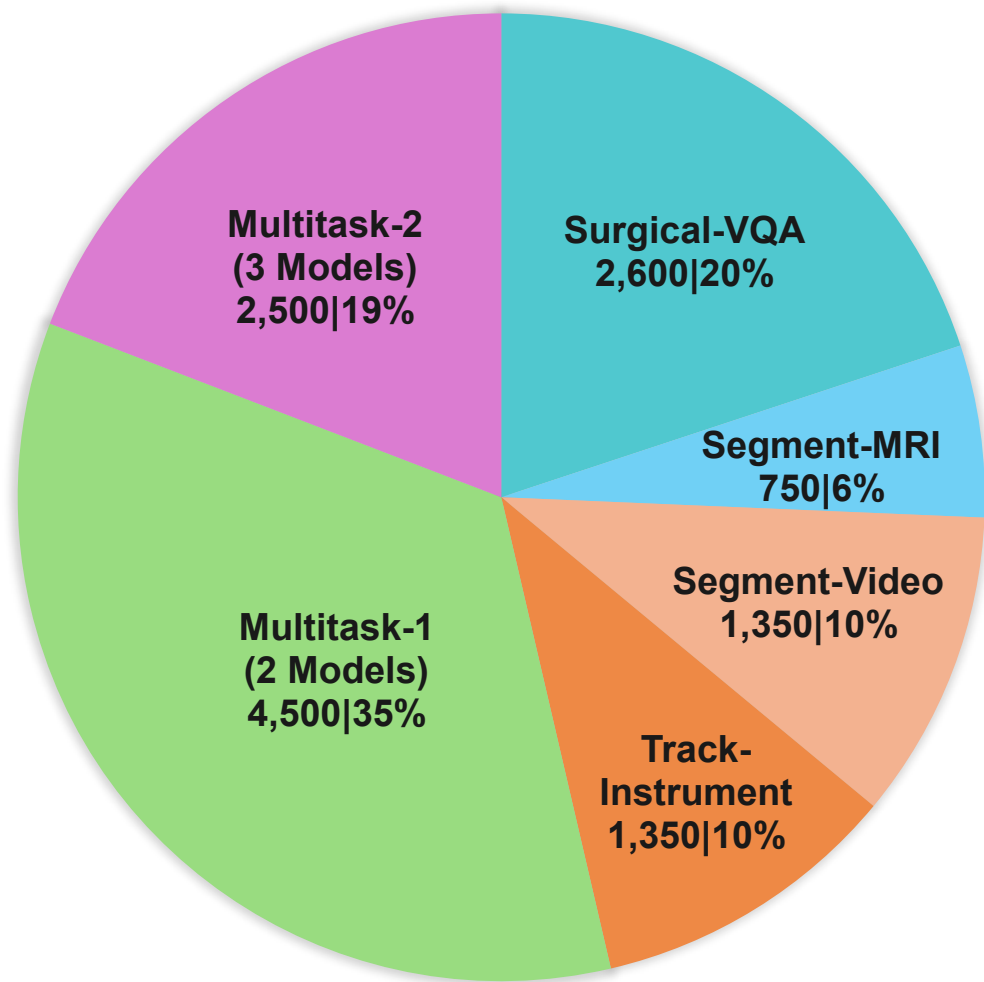
Prompt: Segment right internal carotid artery and sella in the surgical video.

Step 3:

Model: Overlaying

Prompt: Overlay sella and right ICA from MRI onto the surgical video.

(a) A sample annotation of Pit-Agent dataset



(b) Frequency of 6 categories of queries